Prediction of PVT Behaviour, Enthalpy, Entropy and Fugacity of Real Gases using Cubic Equations of State

Upendra Kumar (2025CH11982) & Sadiya Adil (2025CH12131) · Department of Chemical Engineering, IIT Delhi

Web application: *https://upendralatiwal-bit.github.io/Cubic-Equation-of-State-Calculator/*

Source code: *https://github.com/upendralatiwal-bit/Cubic-Equation-of-State-Calculator*

1. Objective

To develop a computer program that (i) predicts the PVT behaviour of pure real gases using cubic equations of state (EOS), (ii) determines the thermodynamic properties enthalpy (H) and entropy (S) from the same EOS through residual properties, and (iii) computes the pure-component fugacity. The program is deployed as an interactive website hosted on GitHub Pages.

2. Theory

2.1 Generic cubic EOS. The van der Waals (vdW), Redlich–Kwong (RK), Soave–Redlich–Kwong (SRK) and Peng–Robinson (PR) equations are all special cases of the generic form (Smith, Van Ness & Abbott):

$$P = RT/(V - b) - a(T)/[(V + \varepsilon b)(V + \sigma b)] \tag{1}$$

$$a(T) = \Psi\, \alpha(T_r)\, R^2 T_c^{\,2}/P_c, \quad b = \Omega\, RT_c/P_c \tag{2}$$

Table 1. Parameters of the cubic equations of state

EOS	$\alpha(T_r)$	σ	ε	Ω	Ψ
vdW	1	0	0	1/8	27/64
RK	$T_r^{-1/2}$	1	0	0.08664	0.42748
SRK	$[1 + (0.480 + 1.574\omega - 0.176\omega^2)(1 - T_r^{1/2})]^2$	1	0	0.08664	0.42748
PR	$[1 + (0.37464 + 1.54226\omega - 0.26992\omega^2)(1 - T_r^{1/2})]^2$	$1 + \sqrt{2}$	$1 - \sqrt{2}$	0.07780	0.45724

With $A = aP/(RT)^2$ and $B = bP/RT$, Eq. (1) becomes a cubic in the compressibility factor $Z = PV/RT$:

$$Z^3 + [(\varepsilon+\sigma)B - 1 - B]Z^2 + [A + \varepsilon\sigma B^2 - (\varepsilon+\sigma)B(1+B)]Z - [AB + \varepsilon\sigma B^2(1+B)] = 0 \tag{3}$$

Below T_c, Eq. (3) may have three real roots: the smallest is liquid-like, the largest vapour-like and the middle one is physically meaningless. Only roots with $Z > B$ are physical.

2.2 Residual enthalpy and entropy. A residual property is $M^R = M - M^{ig}$ at the same T and P. Integrating the EOS gives, with $q = A/B = a/(bRT)$:

$$I = [1/(\sigma - \varepsilon)]\, \ln[(Z + \sigma B)/(Z + \varepsilon B)] \quad (\sigma \neq \varepsilon); \quad I = B/Z \;\; (vdW) \tag{4}$$

$$H^R/RT = Z - 1 + [d\ln\alpha/d\ln T_r - 1]\, qI \tag{5}$$

$$S^R/R = \ln(Z - B) + (d\ln\alpha/d\ln T_r)\, qI \tag{6}$$

where $d\ln\alpha/d\ln T_r = 0$ (vdW), -0.5 (RK) and $-mT_r^{1/2}/[1 + m(1 - T_r^{1/2})]$ for SRK and PR (m is the ω-polynomial in Table 1).

2.3 Total H and S. Taking the ideal gas at $T_0 = 298.15$ K, $P_0 = 1$ bar as the reference ($H = 0$, $S = 0$) and $C_p^{ig}/R = A + BT + CT^2 + DT^{-2}$:

$$H = \int_{T_0}^{T} C_p^{ig}\, dT + H^R \tag{7}$$

$$S = \int_{T_0}^{T} C_p^{ig}\, dT/T - R\, \ln(P/P_0) + S^R \tag{8}$$

2.4 Pure-component fugacity. Since $\ln \varphi = G^R/RT = (H^R - TS^R)/RT$, Eqs. (5) and (6) give

$$\ln \varphi = Z - 1 - \ln(Z - B) - qI, \quad f = \varphi P \tag{9}$$

When three roots exist, the stable phase is the one with the lowest Gibbs energy, i.e. the lowest φ. At saturation, $\varphi^l = \varphi^v$.

3. Algorithm and Software

1. Read the component (T_c, P_c, ω, C_p constants from a built-in database of 12 gases, or entered by the user), the EOS, T, P and the reference state.
2. Compute T_r, P_r, $\alpha(T_r)$, a, b, A, B and the coefficients of Eq. (3).
3. Solve the cubic analytically (Cardano's method; trigonometric form when there are three real roots) and keep the roots with $Z > B$.
4. For each root, compute $V = ZRT/P$, I, H^R, S^R, H, S, φ and f (Eqs. 4–9).
5. Select the stable root (minimum $\ln \varphi$) and identify the phase.
6. Generate plots: P–V isotherms, Z vs P, H^R and S^R vs P, f and φ vs P, and a comparison of all four EOS.

The program is written in JavaScript. The calculation engine is in **eos.js** and the user interface and plots (Chart.js) are in **index.html**. Because it is a static site, it runs entirely in the browser and is hosted free of cost on GitHub Pages. The same engine can be run from the command line with Node.js. All units are SI except P (bar) and V (cm^3/mol); $R = 8.314$ J mol^{-1} K^{-1}.

4. Results and Discussion

Table 2. Program output for the stable root (H, S relative to the ideal gas at 298.15 K, 1 bar)

Gas	EOS	Z	V (cm^3/mol)	H^R (J/mol)	S^R (J/mol·K)	H (J/mol)	S (J/mol·K)	φ	f (bar)
Methane	vdW	0.9005	449.2	-761	-1.710	-696	-34.02	0.9053	45.26
Methane	RK	0.9167	457.3	-799	-1.941	-734	-34.25	0.9168	45.84
Methane	SRK	0.9239	460.9	-832	-2.108	-767	-34.42	0.9232	46.16
Methane	PR	0.9018	449.9	-903	-2.146	-838	-34.45	0.9014	45.07
Carbon dioxide	vdW	0.7313	364.8	-1672	-3.640	-1603	-35.93	0.7925	39.62
Carbon dioxide	RK	0.6972	347.8	-2299	-5.456	-2230	-37.75	0.7668	38.34
Carbon dioxide	SRK	0.6943	346.4	-2644	-6.591	-2575	-38.89	0.7654	38.27
Carbon dioxide	PR	0.6708	334.6	-2708	-6.596	-2639	-38.89	0.7464	37.32
n-Butane	vdW	0.8541	1420.2	-1584	-2.022	23274	33.85	0.8712	21.78
n-Butane	RK	0.8520	1416.7	-1951	-2.709	22907	33.16	0.8664	21.66
n-Butane	SRK	0.8654	1438.9	-2147	-3.199	22711	32.67	0.8766	21.92
n-Butane	PR	0.8463	1407.3	-2245	-3.227	22612	32.64	0.8591	21.48

Conditions: methane and CO_2 at 300 K, 50 bar; n-butane at 500 K, 25 bar.

Table 3. Three roots for n-butane at 350 K and 9.457 bar (PR EOS)

Root	Z	V (cm^3/mol)	H^R (J/mol)	S^R (J/mol·K)	φ	f (bar)
Liquid	0.0366	112.6	-19074	-53.029	0.8383	7.928
Middle (unstable)	0.1318	405.5	-8207	-24.407	1.1220	10.611
Vapour	0.8081	2486.4	-1604	-3.107	0.8374	7.920

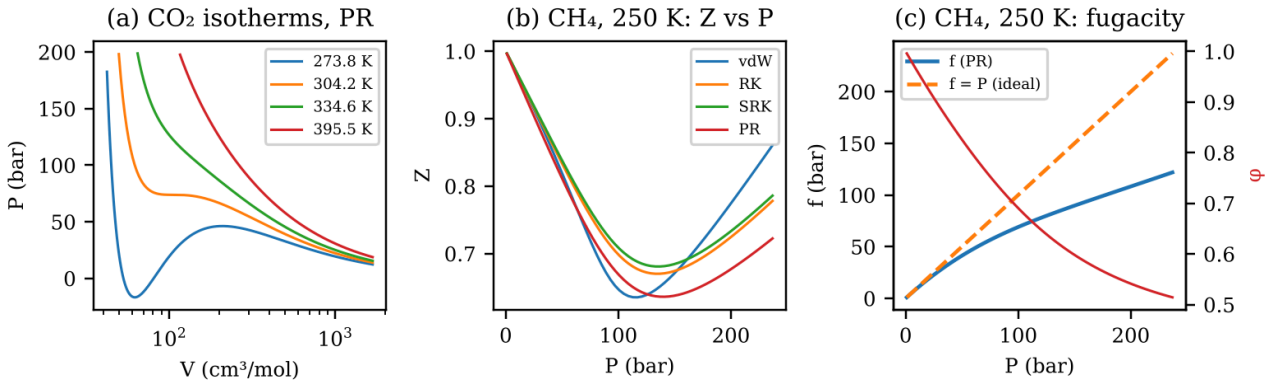


Figure 1. (a) PR isotherms for CO_2 at 0.9, 1.0, 1.1 and 1.3 T_c; (b) Z vs P for methane at 250 K by all four EOS; (c) fugacity and fugacity coefficient of methane at 250 K (PR).

PVT behaviour. For methane at 300 K and 50 bar (T_r = 1.57), all models predict Z = 0.90–0.92, a mild deviation from ideality. The PR value (0.902) is close to the generalized-correlation value of about 0.90. For CO_2 at 300 K ($T_r \approx$ 0.99, near critical), non-ideality is much stronger (Z ≈ 0.67–0.73), and the models differ by up to 9%. The two-parameter vdW equation deviates most because it has no ω or temperature dependence in a. Figure 1(a) reproduces the classic behaviour: van der Waals loops below T_c, an inflection at the critical isotherm and monotonic hyperbola-like isotherms above T_c. Figure 1(b) shows Z first falling with P (attractive forces dominate) and then rising (repulsion dominates).

Enthalpy and entropy. H^R and S^R are negative in every case because the real gas has lower energy and is more ordered than the ideal gas. Their magnitudes grow with the degree of non-ideality: methane at T_r = 1.57 gives $H^R \approx$ −0.9 kJ/mol, whereas near-critical CO_2 gives about −2.7 kJ/mol. Including the temperature derivative of α (SRK, PR) makes H^R more negative than vdW, which has d ln α/d ln T_r = 0 and so underestimates H^R. Total H and S follow from the ideal-gas C_p integral plus the residual correction. For n-butane at 500 K, the ideal-gas sensible heat (≈ 24.9 kJ/mol) dominates and the residual correction is about 9%.

Fugacity. Since φ < 1 in all tabulated cases, attractive forces reduce the "escaping tendency" so that f < P. Fig. 1(c) shows f diverging below the ideal line f = P as pressure rises. Table 3 validates the program: at 350 K, the PR EOS gives three roots for n-butane, and at 9.457 bar the liquid and vapour roots have equal fugacity (7.928 vs 7.920 bar, φ ≈ 0.838). This is the vapour–liquid equilibrium criterion, so 9.46 bar is the PR-predicted vapour pressure, in good agreement with the experimental value of about 9.4 bar. The middle root gives φ > 1 and is rejected.

5. Conclusions

- A single generic algorithm handles all four cubic EOS by changing only (σ, ε, Ω, Ψ, α).
- SRK and PR, which include ω and a temperature-dependent α, give the most realistic PVT, H, S and fugacity predictions, especially near the critical point. The vdW and RK equations are adequate only at high T_r.
- The residual-property route gives H, S and φ consistently from the same EOS. The equal-fugacity condition between liquid and vapour roots correctly locates the vapour pressure.
- The software is freely accessible as a GitHub-hosted web application with interactive plots and editable component data.

References

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